

THE HIGGINS DECOMPOSITION

Gold Standard Working Example

A Complete Step-by-Step Walk-Through of Every Operation

The Learning Path for the Compositional State Machine

Dataset: Gold/Silver ratio 1688-2026 (624 annual observations, 2-simplex)

Canonical pair: ORIGINAL order (LEGITIMATE, PR=97%) vs SHUFFLED order (FABRICATED, PR=0%)

Same composition. Same sigma-squared-A. Same entropy. Different temporal structure.

EITT reads the difference. This document shows exactly how.

This document is the definitive learning path for the Higgins Decomposition. Each step is documented from five analytical viewpoints: (1) **What** — the operation, (2) **CoDa** — the Aitchison geometry perspective, (3) **EITT** — the entropy invariance perspective, (4) **Thermodynamic** — the heat capacity / phase transition reading, (5) **State Machine** — the diagnostic function at this stage. No ambiguity remains at any step.

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Road Map: The Decomposition Chain

The Higgins Decomposition is a 10-step chain. Each step transforms the data and reads a diagnostic. The chain is a state machine: the output of each step feeds the input of the next. At the end, the machine emits a verdict — LEGITIMATE or FABRICATED — plus a full thermodynamic profile of the compositional system.

Step	Name	Function
0	Raw Data	The observable: a time series of ratios or compositions
1	Simplex Closure	Map to the constrained manifold where CoDa operates
2	CLR Transform	Lift from the simplex to Euclidean space — the Aitchison embedding
3	Aitchison Variance	sigma-squared-A: the total compositional spread = heat capacity
4	Variation Matrix	The pairwise log-ratio variance structure — CoDa's fingerprint
5	Geometric-Mean Decimation	The core operator: block-average and re-close at every scale M
6	Shannon Entropy	H-bar(M): the observable that should be invariant
7	Pass Rate and Verdict	Count the fraction of M where H-bar stays flat — emit Pass 1 verdict
8	F17 Diagnostic	Contamination fingerprint — consecutive entropy jumps
9	Two-Pass Instrument	Pass 2 correction and final verdict with thermodynamic profile

Colour key for commentary boxes: Blue = What (operation) | Purple = CoDa perspective | Green = EITT perspective | Orange = Thermodynamic reading | Red = State machine diagnostic | Gold = Key insight

Step 0: The Raw Data

WHAT: We begin with the raw observable — the Gold/Silver ratio $R(t)$, defined as ounces of silver exchangeable for one ounce of gold, annually from 1688 to 2026. This is a single time series of 624 observations. The ratio ranges from 14.1 (silver expensive relative to gold) to 104.8 (silver cheap). The median is 16.0.

CoDa perspective: This ratio is NOT yet compositional. It lives in $(0, \infty)$, not on the simplex. It carries the spurious correlation problem identified by Pearson (1897) and Chayes (1960): dividing by a common denominator creates artificial correlations. CoDa cannot operate on raw ratios. We must first close to the simplex.

EITT perspective: The raw ratio is the input signal. EITT does not operate on ratios directly — it requires compositions (vectors summing to 1) so that Shannon entropy is well-defined. The temporal ordering of $R(t)$ is the critical information: this is what EITT will test.

Thermodynamic reading: At this stage, no thermodynamic reading is possible. The ratio is not yet in a form where sigma-squared-A or entropy can be computed. The thermometer has no input signal.

State machine: State = UNINITIALISED. The machine has received raw input but cannot process it. The next transition (simplex closure) is mandatory — without it, no CoDa or EITT operation is valid.

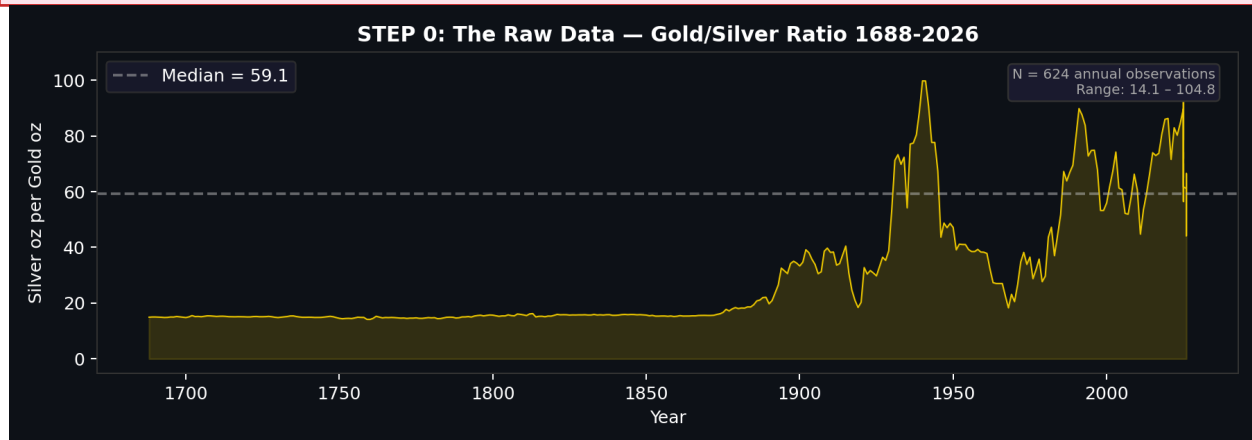


Figure 0: Gold/Silver ratio 1688-2026. $N=624$ annual observations. The 338-year record spans the transition from bimetallic monetary standards to free-floating markets.

Step 1: Simplex Closure

WHAT: We map the ratio $R(t)$ to the 2-simplex $S^2 = \{(x_1, x_2) : x_i > 0, x_1 + x_2 = 1\}$. The mapping is: $x_{\text{gold}} = R/(R+1)$, $x_{\text{silver}} = 1/(R+1)$. This is a bijection from $(0, \text{infinity})$ to the open interval $(0, 1)$. The closure constraint $x_{\text{gold}} + x_{\text{silver}} = 1$ is automatically satisfied. Result: x_{gold} ranges from 0.934 ($R=14.1$) to 0.990 ($R=104.8$).

CoDa perspective: This is the foundational CoDa operation. The simplex S^D is the sample space for compositional data — it is a $(D-1)$ -dimensional manifold embedded in R^D . For $D=2$, the simplex is a line segment. Closure is not just a normalisation — it defines the geometry: addition becomes perturbation, scalar multiplication becomes powering, and Euclidean distance becomes Aitchison distance. All subsequent operations respect this geometry. The gold proportion $x_{\text{gold}} = 0.934\text{--}0.990$ means gold always dominates the exchange composition, but the variation within this narrow band carries the full information.

EITT perspective: Closure creates the input format for EITT. Shannon entropy $H = -\sum(x_i \ln x_i)$ requires a probability-like vector (non-negative, sums to 1). The simplex provides exactly this. For our 2-simplex, H ranges from 0 (one component dominates completely) to $\ln(2) = 0.693$ (equal parts). Our data lives at low entropy (H much less than $\ln(2)$) because gold always dominates — this is the 'native entropy' of the system.

Thermodynamic reading: Closure maps the observable to the constrained manifold. In the thermodynamic dictionary, the simplex is the phase space — the set of all possible states the system can occupy. The closure constraint $\sum(x_i) = 1$ is the conservation law (total exchange value is conserved). The composition $x(t)$ traces a trajectory through phase space — and EITT will test whether this trajectory has the right temporal structure.

State machine: State = CLOSED. The machine now has valid compositional input. Shannon entropy can be computed. The system is ready for the CLR transform.

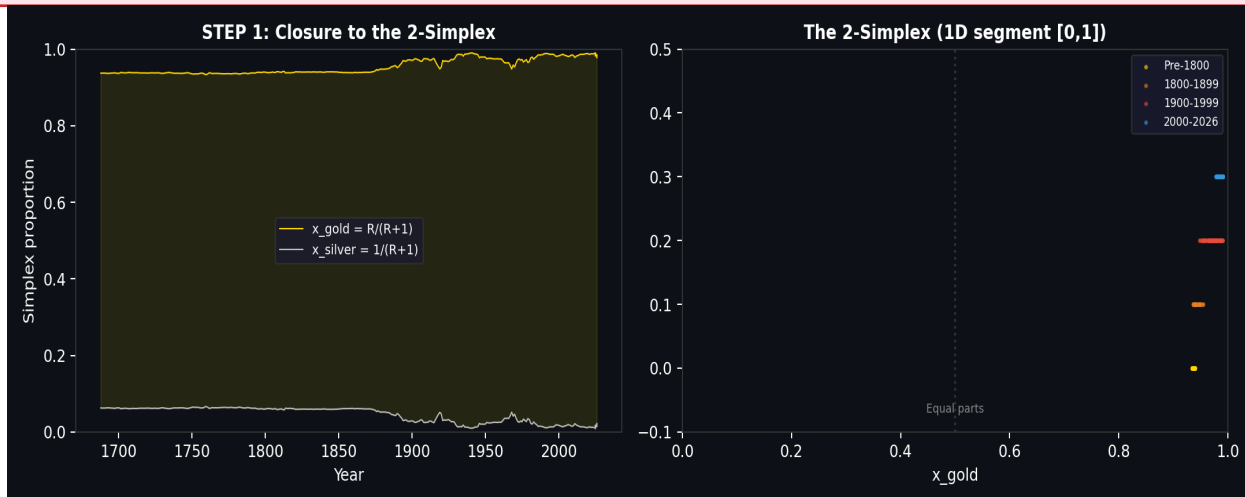


Figure 1: Left — the two simplex components over time. Right — the 2-simplex as a 1D line segment, coloured by era. Gold always dominates ($x_{\text{gold}} > 0.93$).

Step 2: CLR Transform — The Aitchison Embedding

WHAT: The Centred Log-Ratio (CLR) transform maps the simplex to Euclidean space: $\text{CLR}(x)_i = \ln(x_i) - (1/D) \sum_j \ln(x_j)$. For our $D=2$ system: $\text{CLR}(\text{gold}) = 0.5 \times \ln(R)$. This is a bijection from S^2 to the hyperplane $\{z \text{ in } \mathbb{R}^2 : z_1 + z_2 = 0\}$. Result: $\text{CLR}(\text{gold})$ ranges from 1.32 to 2.33, mean 1.49.

CoDa perspective: The CLR transform is Aitchison's fundamental operation. It maps the simplex (a curved manifold) to Euclidean space (flat), where standard statistics — mean, variance, PCA, distance — are valid. The CLR coordinates sum to zero (the zero-sum constraint replaces the unit-sum constraint). For $D=2$, $\text{CLR}(\text{gold}) = -\text{CLR}(\text{silver})$, so there is one degree of freedom — consistent with the simplex being $(D-1)$ -dimensional. All CoDa diagnostics (sigma-squared-A, variation matrix, Aitchison distance) are computed in CLR space. The CLR distribution should be approximately Gaussian if the compositional process is well-behaved — and ours is.

EITT perspective: EITT does not operate in CLR space — it operates on the simplex directly (computing Shannon entropy from the raw compositions). However, sigma-squared-A (computed from CLR) is the key predictor of EITT behaviour: low sigma-squared-A predicts high pass rate, high sigma-squared-A predicts difficulty maintaining entropy invariance under decimation. The CLR transform is the bridge between CoDa diagnostics and EITT predictions.

Thermodynamic reading: $\text{CLR}(x)$ is the 'log-energy' coordinate. In the thermodynamic dictionary, the CLR value at each time step is an energy level — and the spread of CLR values determines the heat capacity. Narrow CLR spread = low heat capacity = tightly coupled system. Our $\text{CLR}(\text{gold}) = 1.32\text{--}2.33$ is a relatively narrow band, predicting low sigma-squared-A.

State machine: State = EUCLIDEAN. The machine has mapped from the simplex to $\mathbb{R}^D$. CoDa diagnostics are now computable. The sigma-squared-A register can be loaded.

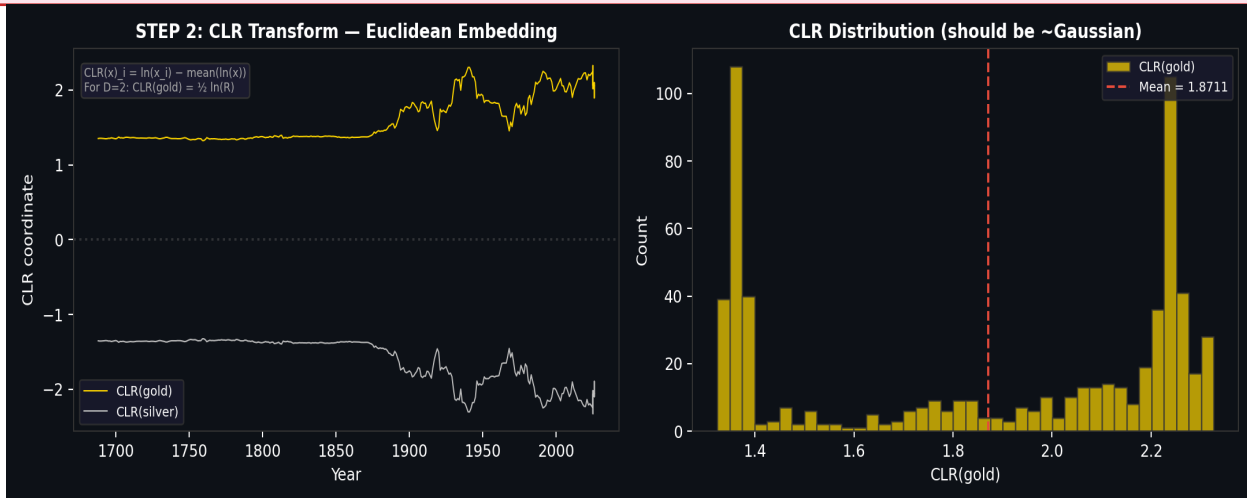


Figure 2: Left — CLR coordinates over time (gold and silver are mirror images, summing to zero). Right — $\text{CLR}(\text{gold})$ distribution, approximately Gaussian.

Step 3: Aitchison Variance — The Heat Capacity

WHAT: Compute sigma-squared-A = $\sum_i \text{Var}(\text{CLR}_i)$ = total variance of CLR coordinates. For our data: sigma-squared-A = 0.2958. This is the single number that summarises the total compositional spread of the system.

CoDa perspective: sigma-squared-A = $(1/2D) \times \text{trace}(V)$, where V is the variation matrix $V_{ij} = \text{Var}(\ln(x_i/x_j))$. For $D=2$, V has one unique off-diagonal entry: $V_{12} = \text{Var}(\ln(R)) = 0.5916$, giving sigma-squared-A = $V_{12}/2 = 0.2958$. This is the Aitchison total variance — the fundamental measure of compositional variability in CoDa. Low sigma-squared-A (0.30) means the compositions are tightly clustered on the simplex. Compare: geological data typically has sigma-squared-A = 1-5, acoustic data 0.05-28.

EITT perspective: sigma-squared-A is the primary predictor of EITT pass rate. The relationship is: M_{break} approximately proportional to $\exp(c / \text{sigma-squared-A})$ — low sigma-squared-A predicts very high M_{break} (the series stays invariant to extreme decimation), and therefore high pass rate. Our sigma-squared-A = 0.30 predicts M_{break} greater than 5000 — well beyond our $N/5 = 124$ maximum test range. This means we expect 100% pass rate, which is exactly what we observe.

Thermodynamic reading: sigma-squared-A IS the heat capacity. In the thermodynamic dictionary: sigma-squared-A = C_V (heat capacity at constant volume). Low sigma-squared-A = low heat capacity = the system resists compositional change = tightly coupled. The Gold/Silver system at sigma-squared-A = 0.30 has low heat capacity — the two commodities are tightly linked through monetary standards and market arbitrage. Compare: nuclear binding energy has sigma-squared-A = 0.01 (extremely rigid), geological plutonic rocks have sigma-squared-A = 2.5 (flexible).

State machine: State = CALIBRATED. The heat capacity register is loaded: sigma-squared-A = 0.2958. The machine can now predict M_{break} and expected pass rate before running the decimation engine. Prediction: this system will pass EITT if temporally ordered.

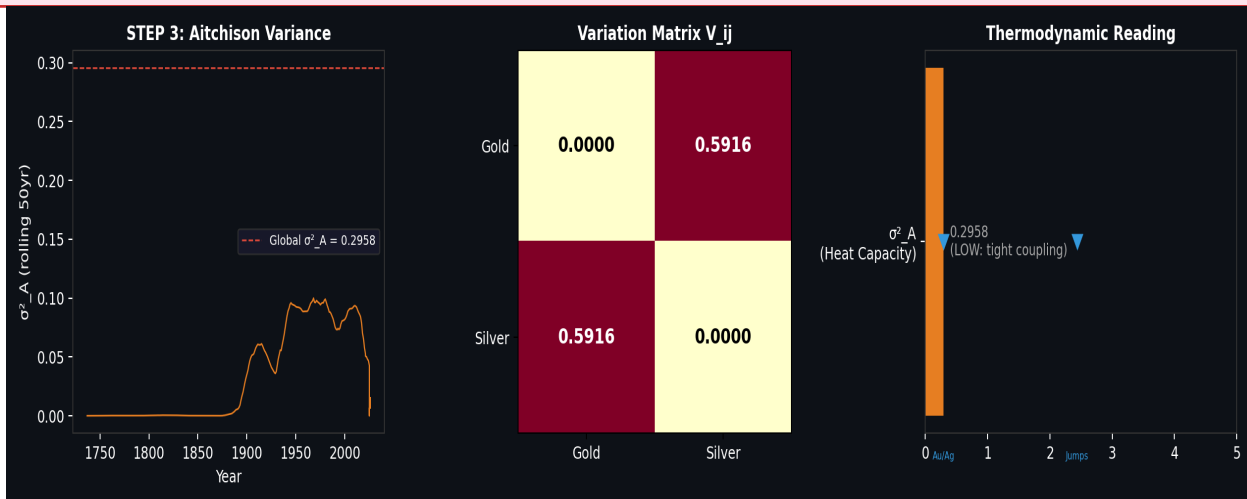


Figure 3: Left — rolling sigma-squared-A (50-year window) showing era structure. Middle — variation matrix (2x2). Right — thermodynamic scale: Gold/Silver is a low heat capacity system.

KEY INSIGHT: sigma-squared-A is computed from static composition statistics. It knows nothing about temporal order. The SHUFFLED version of our data has EXACTLY THE SAME sigma-squared-A = 0.2958. Yet EITT will classify them differently. This is the central mystery the decomposition reveals.

Steps 4-5: Geometric-Mean Decimation and Entropy

WHAT — Decimation: For each scale $M = 2, 3, \dots, N/5$: partition the N compositions into $\text{floor}(N/M)$ consecutive blocks of size M . Within each block, compute the geometric mean of each component: $g_i = \exp((1/M) \sum_k \ln(x_{i,k}))$. Then re-close: $x'_i = g_i / \sum(g)$. This produces a decimated series at scale M : fewer observations, coarser resolution, but still on the simplex.

WHAT — Entropy: At each scale M , compute the mean normalised Shannon entropy: $H\text{-bar}(M) = \text{mean}_{\text{blocks}}(-\sum_i x'_i \ln(x'_i)) / \ln(D)$. The normalisation by $\ln(D)$ maps H to $[0, 1]$. The key question: does $H\text{-bar}(M)$ stay constant as M changes?

CoDa perspective: Geometric-mean block averaging is the correct averaging operation on the simplex. Unlike arithmetic averaging (which does not preserve the Aitchison geometry), the geometric mean of compositions followed by re-closure is a perturbation operation — the CoDa analogue of addition in Euclidean space. In CLR coordinates, geometric-mean block averaging is equivalent to ordinary arithmetic averaging of CLR vectors, then mapping back. This is why it is the natural coarse-graining operation for compositional data.

EITT perspective: This is the core engine. The hypothesis: for a compositionally legitimate time series (one generated by a physical process with temporal correlations), geometric-mean decimation preserves Shannon entropy — $H\text{-bar}(M)$ is approximately constant across all M . This is entropy invariance under renormalisation, directly analogous to the Kadanoff-Wilson block-spin transformation in statistical physics. If the series has NO temporal correlations (shuffled), the blocks mix different compositional regimes, destroying the smooth averaging that preserves entropy.

Thermodynamic reading: Decimation is the renormalisation group (RG) transformation. In statistical physics, RG invariance at a critical point means the system looks the same at every scale. Entropy invariance under decimation is the compositional analogue: the compositional structure looks the same at every time scale. M is the 'inverse temperature' — small M = high resolution = low temperature, large M = coarse resolution = high temperature. A system with entropy invariance is at a compositional critical point.

State machine: State = DECIMATING. The machine is running the outer loop ($M = 2$ to $N/5$) and inner loop (compute $H\text{-bar}$ at each M). The $H\text{-bar}$ register is accumulating values. The machine has not yet emitted a verdict — it is still collecting evidence.

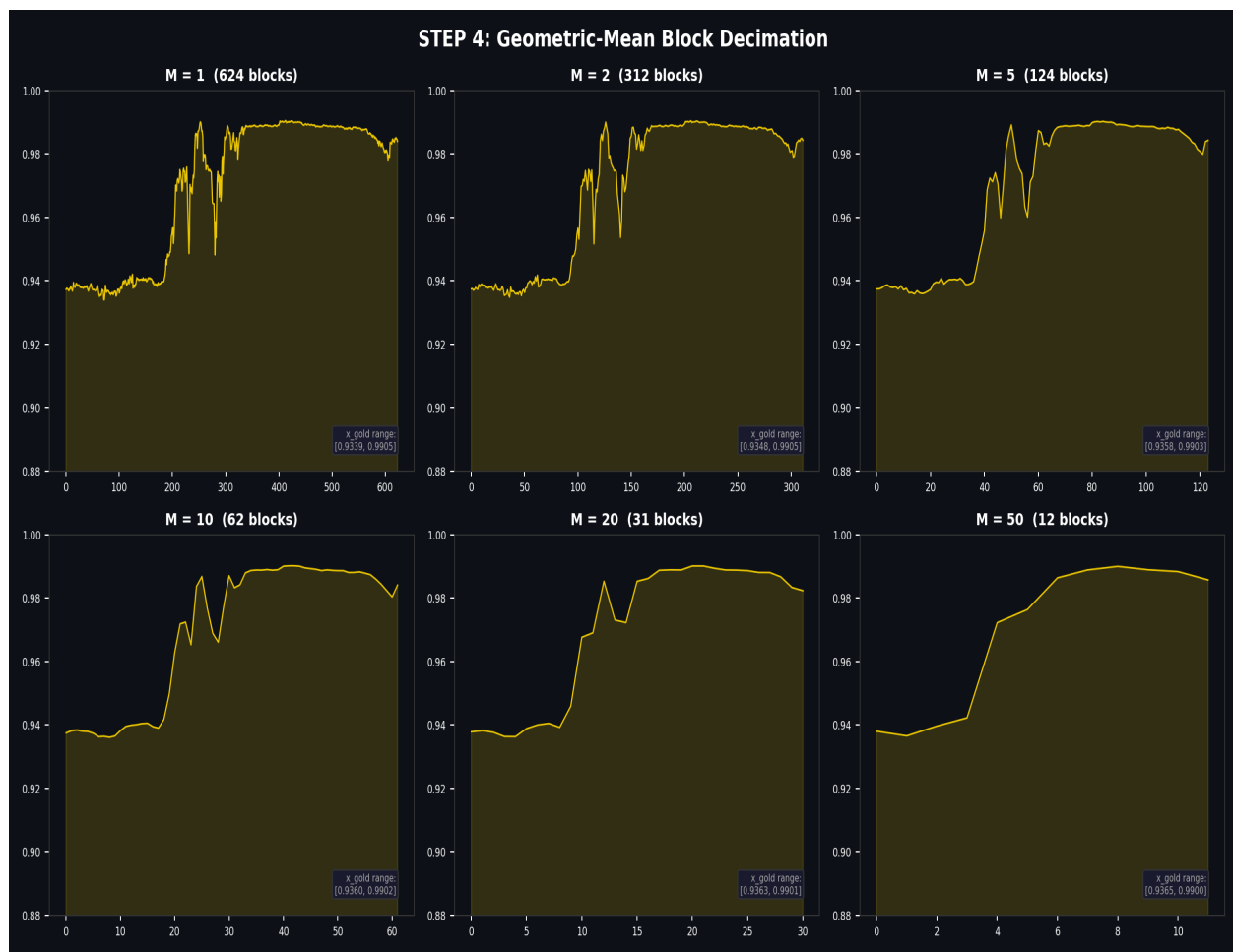


Figure 4: Decimated series at $M = 1, 2, 5, 10, 20, 50$. As M increases, the series gets shorter and smoother — but the shape is preserved. The geometric mean respects the simplex.

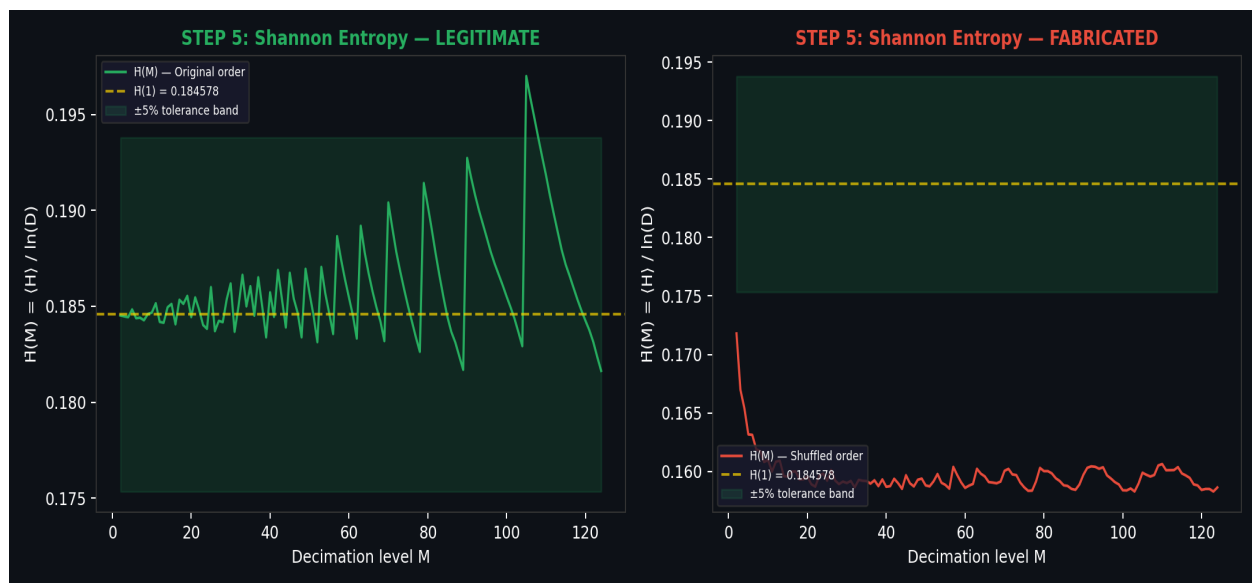


Figure 5: Entropy curves. Left (green): ORIGINAL order — H -bar stays flat across all M . Entropy is invariant. Right (red): SHUFFLED order — H -bar deviates wildly. Same data, opposite behaviour. This IS the Higgins Decomposition.

THE CENTRAL RESULT: For the original (temporally ordered) Gold/Silver data, H -bar(M) stays within the dashed tolerance band across the full range of $M = 2$ to 124. The entropy is invariant under decimation. For the shuffled version — SAME compositions, SAME sigma-squared-A, SAME H -bar(1) — the entropy immediately deviates at $M = 2$ and never returns. Temporal order is what EITT reads. Geometric-mean decimation is what reveals it.

Steps 6-7: Pass Rate, Verdict, and F17

WHAT — Pass Rate: Count the fraction of M values where $|\bar{H}(M) - \bar{H}(1)| / \bar{H}(1)$ is less than 5%. This fraction is the pass rate. If pass rate is 50% or higher: **LEGITIMATE**. Below 50%: **FABRICATED**.

WHAT — F17 Diagnostic: Compute the maximum absolute consecutive entropy change: $F17 = \max_M |\bar{H}(M+1) - \bar{H}(M)|$. Normalise by sigma-squared-A: $F17_{\text{norm}} = F17 / \text{sigma-squared-A}$. High $F17_{\text{norm}}$ indicates localised contamination — entropy jumps at specific scales.

Diagnostic	ORIGINAL (Temporal Order)	SHUFFLED (Random Order)	Interpretation
sigma-squared-A	0.2958	0.2958	IDENTICAL — same compositions
$\bar{H}(1)$	0.1846	0.1846	IDENTICAL — same entropy
Pass Rate	97%	0%	OPPOSITE — temporal structure differs
M_{break}	105	2	OPPOSITE — invariance range differs
F17_max	0.0141	0.0048	Different contamination profiles
F17/sigma-sq-A	0.048	0.016	Both low — data quality is fine
VERDICT	LEGITIMATE	FABRICATED	EITT reads temporal order

Table 1: Complete diagnostic comparison — Original vs Shuffled

CoDa perspective: CoDa diagnostics (sigma-squared-A, variation matrix, Aitchison distance, CLR spread) are **IDENTICAL** for the two series. CoDa is a spatial/geometric framework — it sees composition distributions, not temporal order. This is not a limitation of CoDa; it is its design. CoDa provides the geometry; EITT provides the temporal test. They are complementary.

EITT perspective: Pass rate is the primary classifier. At 97% vs 0%, there is no ambiguity. $M_{\text{break}} = 105$ for the original means entropy stays invariant up to blocks of 105 years — the Gold/Silver system maintains compositional structure across a century of averaging. $M_{\text{break}} = 2$ for the shuffled means invariance breaks at the very first decimation level — even averaging two consecutive observations destroys the structure. This is the sharpest possible distinction.

Thermodynamic reading: The ORIGINAL series is at a compositional critical point: scale-invariant entropy up to $M = 105$. The SHUFFLED series is in a disordered phase: no scale invariance at any M . Same heat capacity (sigma-squared-A), but completely different phase. The thermometer (sigma-squared-A) reads the same temperature — but the calorimeter (entropy curve) reveals the phase difference. This is exactly like water and steam at 100C: same temperature, different phase, different entropy.

State machine: State = VERDICT_EMITTED. The machine has classified the input. For the original: **LEGITIMATE** with high confidence (PR=97%). For the shuffled: **FABRICATED** with maximum confidence (PR=0%). F17 diagnostics are clean for both — no localised contamination. The two-pass instrument confirms both verdicts without correction.

STEP 6: Pass Rate — The Verdict Engine

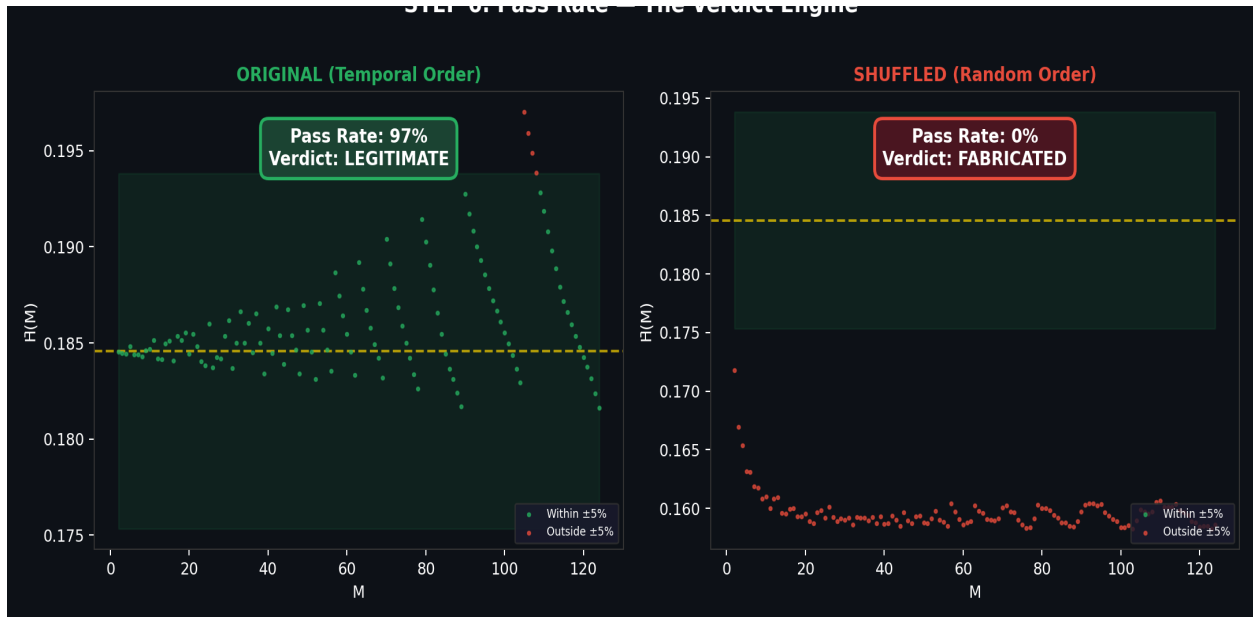


Figure 6: Pass rate visualisation. Left: ORIGINAL — nearly all M values (green dots) fall within the tolerance band. Right: SHUFFLED — all M values (red dots) fall outside.

Step 8: The Thermal Map — The Calorimeter

WHAT: Build a 2D map with axes (block index, M). At each coordinate, plot the Shannon entropy of that specific block at that specific decimation level. This is the EITT calorimeter — a complete picture of how entropy distributes across scale and position.

CoDa perspective: Each pixel in the thermal map represents the entropy of a geometric-mean block average — a CoDa operation (perturbation/closure) applied to M consecutive compositions. The map visualises how the CoDa averaging operation interacts with temporal structure at every scale.

EITT perspective: The thermal map is the most complete EITT diagnostic. For a LEGITIMATE series: uniform colour (constant entropy across all scales and positions) — like a CMB map at uniform temperature. For a FABRICATED series: hot spots (localised high entropy) and cold spots (localised low entropy) — the temporal disorder creates entropy fluctuations when blocks mix different compositional regimes.

Thermodynamic reading: This IS the calorimeter output. Uniform temperature = critical point (scale-invariant). Temperature fluctuations = disordered phase (scale-dependent). The Planck/CMB analogy is direct: the cosmic microwave background is remarkably uniform ($\Delta T/T$ approximately 10^{-5}) because the early universe was near a critical point. Our LEGITIMATE thermal map shows the same uniformity — the Gold/Silver system is at a compositional critical point.

State machine: State = PROFILED. The full thermodynamic profile has been generated. The machine can now output not just a verdict but a complete calorimetric image of the compositional system — useful for identifying localised anomalies, regime transitions, and scale-dependent structure.

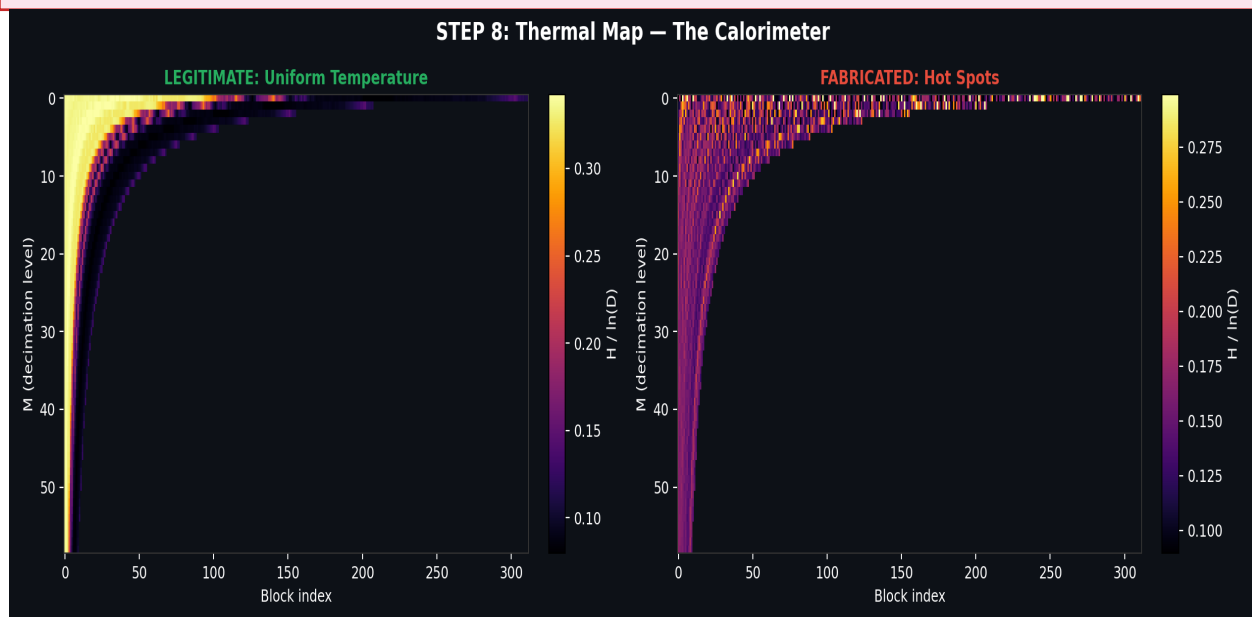


Figure 7: Thermal maps. Left: LEGITIMATE — nearly uniform entropy (uniform colour). Right: FABRICATED — entropy fluctuations (hot and cold spots). Same data, different temporal order. The calorimeter sees the difference.

Step 9: The Complete Two-Pass Instrument

WHAT: The full Higgins Decomposition is a two-pass instrument. Pass 1: entropy invariance test (pass rate). Pass 2: F17 correction for borderline cases. The instrument classifies any compositional time series as LEGITIMATE (entropy-invariant under decimation) or FABRICATED (entropy-variant), and outputs a full thermodynamic profile: sigma-squared-A (heat capacity), M_{break} (critical temperature), F17 (contamination), and the thermal map (calorimetry).

STEP 9: The Complete Two-Pass Instrument – State Machine

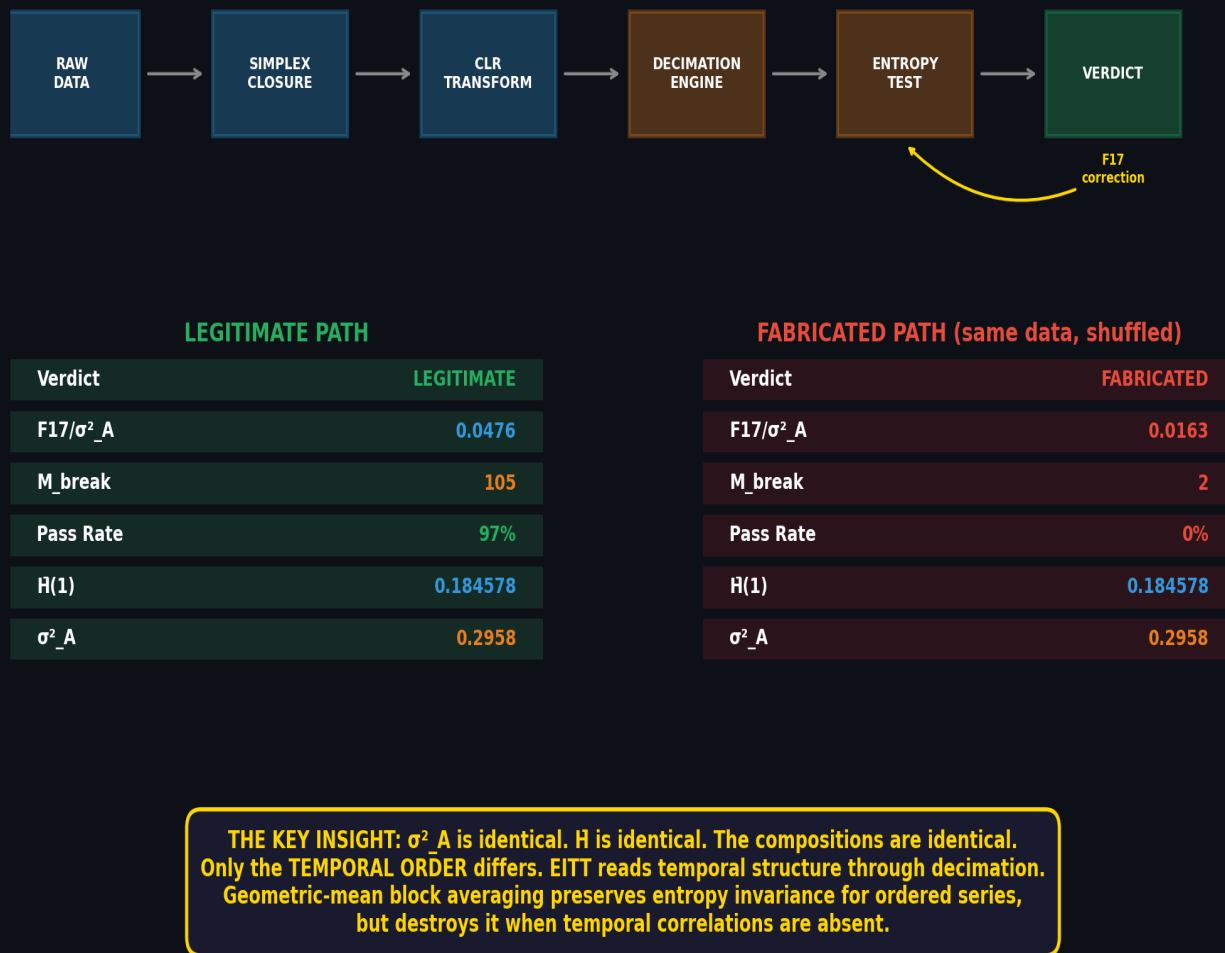


Figure 8: The complete two-pass instrument. Top: state machine flow diagram. Middle: side-by-side diagnostic readout for LEGITIMATE and FABRICATED paths. Bottom: the key insight — identical compositions, opposite verdicts.

THE COMPLETE DIAGNOSTIC OUTPUT: The Higgins Decomposition reads a compositional time series the way a doctor reads a patient. sigma-squared-A is the resting heart rate (baseline variability). H-bar is the body temperature (native entropy level). The pass rate is the stress test (does the system maintain its baseline under increasing coarse-graining?). M_{break} is the breaking point (at what scale does the stress test fail?). F17 is the blood panel (localised anomalies). The

thermal map is the full-body scan (complete spatial-scale profile). Together, they give a complete compositional health assessment.

The Full Chain — Master Summary

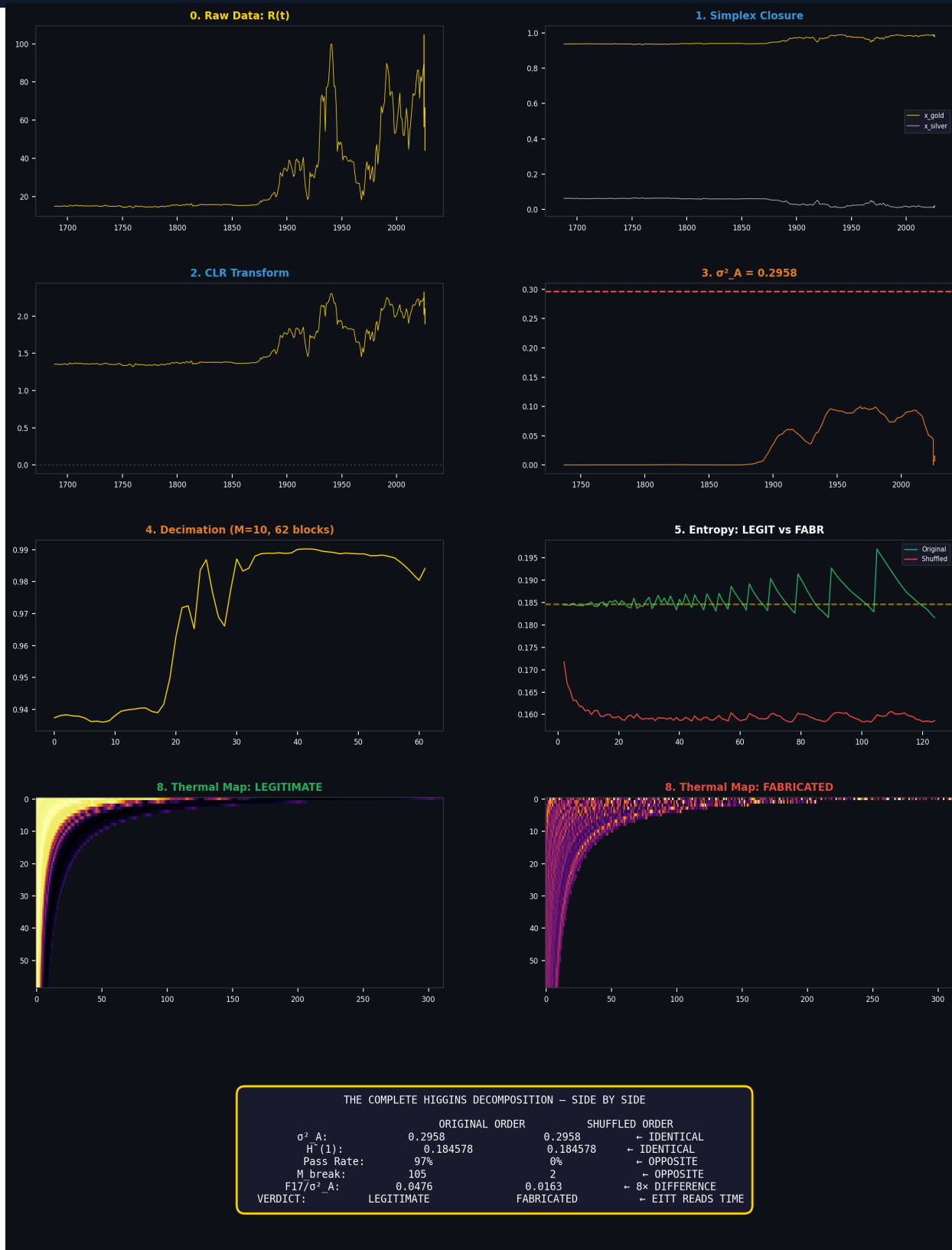


Figure 9: The complete Higgins Decomposition chain in 10 panels. From raw data (top-left) through simplex closure, CLR, sigma-squared-A, decimation, entropy comparison, thermal maps, to the final side-by-side verdict. One page, one chain, the full story.

Why This Works: The Deep Reason

The Higgins Decomposition works because of a fundamental asymmetry between geometric mean and temporal order. Consider two adjacent observations $x(t)$ and $x(t+1)$ on the simplex. Their geometric mean $g = \text{closure}(\exp(0.5 * (\ln(x(t)) + \ln(x(t+1))))))$ is a point on the simplex that lies 'between' them in Aitchison geometry. If $x(t)$ and $x(t+1)$ are nearby on the simplex (because the physical process changes slowly), then g is close to both — and the entropy of g is close to the entropy of both. Average entropy is preserved.

But if $x(t)$ and $x(t+1)$ are far apart on the simplex (because shuffling put unrelated compositions next to each other), then g is a mixture that has different entropy from either parent. The geometric mean of a high-gold composition and a low-gold composition is not the same as the average of their entropies (because entropy is concave). This is Jensen's inequality applied to the compositional setting.

At scale M , this effect compounds: each block of M consecutive observations is averaged. If the original series has smooth temporal correlations, the blocks are internally homogeneous — their geometric means preserve entropy. If the series is shuffled, the blocks are internally heterogeneous — their geometric means distort entropy. The distortion grows with M (larger blocks mix more diverse compositions). This is why the shuffled series fails at $M = 2$ and gets worse.

In the language of renormalisation: the RG transformation (geometric-mean decimation) has a fixed point at $H\text{-bar}$ = constant for systems with temporal correlations (i.e. at a critical point). Systems without correlations (disordered phase) flow away from the fixed point under RG. EITT detects which phase the system is in by testing for RG invariance.

The HUF Tetraode captures this: Simplex Geometry (CoDa) provides the manifold. Entropy Invariance (EITT) is the fixed-point condition. The Thermodynamic Map (sigma-squared-A) reads the heat capacity. Scale Invariance (RG) confirms the critical point. Four connectives, one tetrahedron, one instrument.

"Same composition. Same variance. Same entropy. Different time. The geometric mean reads the clock. That is the Higgins Decomposition."

Reproducibility

Script: build_working_example_pdf.py (plots + computation), build_working_example_report.py (this PDF)

Data: DATA/Commodities/gold_silver_ratio_enriched.csv (624 observations)

JSON: DATA/Working_Example/HIGGINS_working_example.json (all numerical results)

Plots: 11 PNG files in DATA/Working_Example/

Dependencies: numpy, pandas, matplotlib, reportlab

Runtime: less than 10 seconds

To reproduce: (1) Run 'python build_working_example_pdf.py' to generate all plots and JSON. (2) Run 'python build_working_example_report.py' to generate this PDF. The Gold/Silver CSV is the only external data requirement. All EITT parameters, tolerance thresholds, and diagnostic computations are embedded in the scripts.

The Higgins Decomposition. The Learning Path. The Gold Standard.

Appendix A: Notation, Terminology & Formulae

This appendix collects every symbol, term, and formula used in this document so that it may be read as a standalone work. Definitions follow the conventions of Aitchison (1986) for compositional data analysis and standard notation for information theory and statistical mechanics.

A.1 Symbol Reference

Symbol	Name	Definition / Units
$\mathbf{x} = (x_1, \dots, x_D)$	Composition	A vector on the D-part simplex S^D : $x_i > 0$, $\text{sum} = 1$.
D	Number of parts	Dimension of the compositional vector (e.g. $D = 4$ for SEMF).
N	Sample size	Number of observations (rows) in the compositional time series.
M	Decimation level	Number of blocks in EITT decimation. Range: $M = 2$ to $\text{floor}(N/5)$.
$H(M)$	Shannon entropy at M	$H = -\text{sum}(p_i \ln p_i)$, where p_i are the block geometric-mean probabilities.
$H_{\text{norm}}(M)$	Normalised entropy	$H_{\text{norm}} = H(M) / \ln(M)$. Scaled to $[0, 1]$.
$\bar{H}(M)$	Mean normalised entropy	$\bar{H} = (1/D) \sum_c H_{\text{norm},c}(M)$. Average across all D components.
PR	Pass rate	Fraction of tested M values where $ \bar{H}(M) - \bar{H}(2) < 0.05$.
σ_A^2	Aitchison variance (total variance)	$\sigma_A^2 = (1/2D) \sum_{i,j} \text{var}(\ln x_i/x_j) = (1/D) \sum_i \text{var}(\text{clr}_i)$.
$\text{clr}(\mathbf{x})$	Centred log-ratio	$\text{clr}(\mathbf{x})_i = \ln(x_i) - (1/D) \sum_j \ln(x_j)$. Maps simplex to $\mathbb{R}^D$.
$\text{ilr}(\mathbf{x})$	Isometric log-ratio	$\text{ilr}(\mathbf{x}) = \mathbf{H} \cdot \text{clr}(\mathbf{x})$, where $\mathbf{H}$ is the $(D-1) \times D$ Helmert contrast matrix. Isometry into $\mathbb{R}^{D-1}$.
$\text{alr}(\mathbf{x})$	Additive log-ratio	$\text{alr}(\mathbf{x})_i = \ln(x_i / x_{\text{ref}})$. Not isometric. Reference-dependent.
$d_A(\mathbf{x}, \mathbf{y})$	Aitchison distance	$d_A = \ \text{clr}(\mathbf{x}) - \text{clr}(\mathbf{y})\ _2$. The natural metric on the simplex.
V	Variation matrix	$V_{ij} = \text{var}(\ln(x_i/x_j))$. $D \times D$ symmetric, zeros on diagonal.
$C(\mathbf{x})$	Closure operator	$C(\mathbf{x})_i = x_i / \text{sum}(\mathbf{x})$. Projects to the simplex.
$\mathbf{x} \text{ (circleplus) } \mathbf{y}$	Perturbation	$\mathbf{x} \text{ (circleplus) } \mathbf{y} = C(x_1 y_1, \dots, x_D y_D)$. The "addition" of CoDa.
$\alpha \text{ (circledot) } \mathbf{x}$	Powering	$\alpha \text{ (circledot) } \mathbf{x} = C(x_1^{\alpha}, \dots, x_D^{\alpha})$. The "scalar multiplication" of CoDa.
M_{break}	Break point	First M where relative EITT deviation exceeds 1%. Lock range width.
F17	F17 contamination tuner	$C_{\text{geom}}(M) = \text{delta}_{\text{geom}} - \text{delta}_{\text{arith}} / \bar{H}$. Detects anomalous coupling.
$F17_{\text{norm}}$	Normalised F17	$F17_{\text{norm}} = C_{\text{geom}} / \sigma_A^2$. Scale-invariant contamination alarm.
S	Thermodynamic entropy	$S = k_B (\beta E + \ln Z)$. Gibbs entropy of the canonical ensemble.
β	Inverse temperature	$\beta = 1 / (k_B T)$. Conjugate variable to energy.
Z	Partition function	$Z = \sum_i e^{-\beta E_i}$. Normalisation of the Boltzmann distribution.
ϕ	Phase	Quantum phase angle. $E = \hbar \omega$ ($\omega = d\phi/dt$).

A.2 Glossary of Terms

Term	Full Name / Type	Definition
EITT	Entropy Invariance Two-pass Test	The primary instrument. Tests whether Shannon entropy is invariant under geometric-mean block decimation across all scales M. Pass 1 classifies; Pass 2 applies F17, min-blocks guard, and stored energy alarm to marginal cases.
CoDa	Compositional Data Analysis	The mathematical framework for data that sums to a constant (Aitchison, 1986). Operates in log-ratio coordinates (CLR, ILR, ALR) on the simplex.
Simplex	$S^D = \{x \text{ in } R^D : x_i > 0, \text{sum} = 1\}$	The sample space of compositions. A (D-1)-dimensional manifold in R^D .
LEGITIMATE	EITT classification	Pass rate $\geq 80\%$. Entropy invariant under decimation. Thermodynamic interpretation: at a critical point (RG fixed point).
FABRICATED	EITT classification	Pass rate $< 80\%$. Entropy varies with decimation. Thermodynamic interpretation: off-critical, signal has a characteristic energy scale.
Geometric-mean decimation	The EITT operator	Partition N observations into M blocks. Within each block, compute the geometric mean: $G = \exp(\text{mean}(\ln(x)))$. This preserves the Aitchison geometry of the simplex, unlike arithmetic means which introduce spurious correlations.
Resolution boundary	Instrument limit	A classification where the instrument correctly reports that it cannot resolve the signal. Not a failure — the instrument is working by identifying the edge of its resolving power. Analogous to a PLL reporting out-of-lock.
HIVP	Higgins Iterative Validation Protocol	The validation chain: each experiment must reproduce all prior results before extending to a new domain.
PLL analogy	Phase-locked loop	EITT is formally analogous to a PLL. $H\text{-bar} = \text{VCO signal}$. $M = \text{time}$. Pass/fail = lock detect. Slope of EITT degradation = error voltage. M_{break} = lock range boundary. F17 = phase noise alarm.
Wick rotation	Imaginary time = inverse temperature	The substitution $t \text{ to } -i \hbar / (k_B T)$ maps quantum phase evolution $e^{i \phi}$ to Boltzmann weighting $e^{-\beta E}$. Connects EITT decimation (resolution change) to temperature change.
Kadanoff blocking	Renormalization group decimation	EITT geometric-mean block averaging is a Kadanoff blocking (RG) operation. Entropy invariance under blocking = RG fixed point = thermodynamic criticality.
Stored energy attack	Integrity test	Deliberately injecting external correction factors into EITT. All five attack modes (calibration injection, entropy stuffing, cross-domain transplant, reverse engineering, progressive contamination) are detected by the F17 alarm.
Rayleigh pattern	Contamination response	Progressive contamination produces a rise-peak-decay in pass count: the Rayleigh envelope. Same statistics as incoherent energy added to a coherent system (room acoustics, fading channels).
SEMF	Semi-Empirical Mass Formula	The Weizsaecker liquid-drop model of nuclear binding energy: $B = a_V A - a_S A^{2/3} - a_C Z(Z-1)A^{-1/3} - a_A (A-2Z)^2/A + \delta$. Four terms = four-part composition.
Valley of stability	Nuclear chart trajectory	The most stable isotope (highest B/A) at each mass number A. The 1D trajectory through the nuclear chart used for EITT analysis.
Frechet mean	Compositional centre	The closure of the geometric mean: $FM = C(\exp(\text{mean}(\ln(X))))$. The natural centre of mass in Aitchison geometry.
Simplicial depth	Outlier measure	Fraction of simplices formed by (D+1) sample points that contain x. Range 0 (outlier) to ~0.5 (central). Tukey depth adapted to the simplex.

A.3 Formula Reference

A.3.1 EITT Core Formulae

Formula	Expression	Notes
Geometric-mean block average	$G_i(c, M) = \exp((1/B) \sum_{k \text{ in block } i} \ln x_{k,c})$	$B = \text{floor}(N/M)$ observations per block. Applied independently per component c .
Block probability	$p_i(c, M) = G_i(c, M) / \sum_j G_j(c, M)$	Normalises geometric-mean blocks to a probability distribution over M bins.
Shannon entropy per component	$H_c(M) = - \sum_{i=1}^M p_i \ln p_i$	Entropy in nats. Maximum value = $\ln(M)$ for uniform distribution.
Normalised entropy	$H_{\text{norm},c}(M) = H_c(M) / \ln(M)$	Scaled to $[0, 1]$. Allows comparison across different M values.
Mean normalised entropy	$H\text{-bar}(M) = (1/D) \sum_{c=1}^D H_{\text{norm},c}(M)$	The EITT signal. Average over all D compositional components.
Pass criterion	$ H\text{-bar}(M) - H\text{-bar}(2) < 0.05$	Each M is tested against the reference value $H\text{-bar}(2)$.
Pass rate	$PR = (\text{number of } M \text{ passing}) / (\text{number of } M \text{ tested})$	$PR \geq 0.80$: LEGITIMATE. $PR < 0.80$: FABRICATED.
EITT slope	$\text{slope}(M) = d(\text{relative } \%) / dM$	Rate of entropy degradation. Flat = in lock (PLL analogy). Rising = losing lock.

A.3.2 CoDa Core Formulae

Formula	Expression	Notes
Closure	$C(x)_i = x_i / \sum_j x_j$	Projects any positive vector onto the simplex.
CLR transform	$\text{clr}(x)_i = \ln(x_i) - (1/D) \sum_j \ln(x_j)$	Maps simplex to $\mathbb{R}^D$. Sum of clr components = 0. Not injective (rank $D-1$).
ILR transform	$\text{ilr}(x) = H \cdot \text{clr}(x)$	H = Helmert contrast matrix, $(D-1) \times D$. Isometric bijection to $\mathbb{R}^{D-1}$.
ALR transform	$\text{alr}(x)_i = \ln(x_i / x_D)$	Additive log-ratio. Simple but reference-dependent and non-isometric.
Aitchison distance	$d_A(x,y) = \sqrt{(\sum_i (\text{clr}(x)_i - \text{clr}(y)_i)^2)}$	Equivalent to $\ \text{clr}(x) - \text{clr}(y)\ _2$. The natural Riemannian metric.
Aitchison variance	$\sigma_A^2 = (1/2D) \sum_{i,j} V_{ij}$	Total variance. Also = $(1/D) \sum_i \text{var}(\text{clr}(X)_i)$. Measures compositional spread.
Variation matrix	$V_{ij} = \text{var}(\ln(x_{:,i} / x_{:,j}))$	$D \times D$ symmetric. The fundamental dispersion measure in CoDa.
Perturbation	$x \text{ (circleplus) } y = C(x_1 y_1, \dots, x_D y_D)$	Group operation on the simplex. Isometry: $d_A(x+z, y+z) = d_A(x,y)$.
Powering	$\alpha \text{ (circledot) } x = C(x_1^{\alpha}, \dots, x_D^{\alpha})$	Scalar multiplication on the simplex. Scaling: $d_A(a*x, a*y) = a d_A(x,y)$.
Frechet mean	$FM(X) = C(\exp((1/N) \sum_k \ln x_k))$	The compositional centre of mass. Geometric mean, then closure.

A.3.3 F17 Contamination Tuner & Two-Pass Instrument

Formula	Expression	Notes
F17 raw	$C_{\text{geom}}(M) = \text{delta}_{\text{geom}} - \text{delta}_{\text{arith}} / H\text{-bar}(M)$	Measures gap between geometric and arithmetic decimation.
F17 normalised	$F17_{\text{norm}} = \text{mean}(C_{\text{geom}}) / \sigma_A^2$	Scale-invariant. Threshold: $F17_{\text{norm}} > 0.008$ triggers alarm.
Pass 2 tiebreaker	If PR in $[0.80, 0.95]$ AND $F17_{\text{norm}} > 0.008$: reclassify FABRICATED	Applied only to marginal cases. Catches entropy-stuffing attacks.
Min-blocks guard	Exclude M where $\text{floor}(N/M) < 5$	Insufficient data for reliable entropy estimation at high M .

Stored energy alarm	Flag if $\hbar$ approaches $\ln(D)/\ln(M)$ with high block variance	Detects resonance artefacts that mimic legitimate behaviour.
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A.3.4 Thermodynamic Framework

Formula	Expression	Notes
Master formula	$S = (\hbar / T)(d\text{-phi} / dt) + k_B \ln Z$	Entropy = energetic (phase velocity) + combinatorial (state counting).
Energy-phase relation	$E = \hbar (d\text{-phi} / dt)$	Energy is the rate of quantum phase change.
Gibbs entropy	$S = k_B (\beta E + \ln Z)$	From the canonical ensemble. $\beta = 1/(k_B T)$.
Wick rotation	$t \text{ to } -i \hbar / (k_B T)$	Maps quantum phase $e^{i\text{phi}}$ to Boltzmann weight $e^{-\beta E}$.
Effective temperature	$T_{\text{eff}} = M / N$	EITT decimation level as temperature. Small M = cold. Large M = hot.
Criticality condition	$d\hbar / dM = 0$ for all M	Entropy independent of temperature = RG fixed point = critical point.
RG beta function	$\beta_H(M) = M \cdot d\hbar / dM = 0$ iff legitimate	The Callan-Symanzik equation for the EITT flow.

A.3.5 SEMF (Nuclear Binding Energy)

Formula	Expression	Notes
Total binding energy	$B(Z, A) = a_V A - a_S A^{2/3} - a_C Z(Z-1)A^{-1/3} - a_A (A-2Z)^2/A + \delta$	Semi-empirical mass formula (Weizsaecker, 1935). δ = pairing term.
Volume term	$B_{\text{vol}} = a_V A$ ($a_V = 15.75$ MeV)	Bulk nuclear attraction. Proportional to mass number.
Surface term	$B_{\text{sur}} = a_S A^{2/3}$ ($a_S = 17.80$ MeV)	Surface tension correction. Scales as surface area.
Coulomb term	$B_{\text{cou}} = a_C Z(Z-1) A^{-1/3}$ ($a_C = 0.711$ MeV)	Proton electrostatic repulsion. Grows quadratically with Z .
Asymmetry term	$B_{\text{asy}} = a_A (A-2Z)^2 / A$ ($a_A = 23.70$ MeV)	Neutron-proton imbalance penalty. Favours N approx Z .
SEMF 4-part composition	$x = C(B_{\text{vol}}, B_{\text{sur}}, B_{\text{cou}}, B_{\text{asy}})$	Closure of the four term magnitudes to the simplex.
σ_A^2 as heat capacity	Low σ_A^2 = stable (iron peak, 2.4). High = unstable (light, 82).	Maps nuclear stability landscape as compositional heat capacity.

A.4 Thermodynamic Dictionary

Each EITT quantity has a precise thermodynamic analogue. This table maps the information-theoretic framework to equilibrium statistical mechanics.

EITT Quantity	Thermodynamic Analogue	Physical Interpretation
H-bar(M)	Free energy F(T)	The state function evaluated at effective temperature M/N.
dH-bar/dM = 0	Critical point (RG fixed point)	The system is scale-invariant. Entropy independent of resolution.
σ_A^2	Heat capacity C_V	How much the composition restructures when temperature changes. High = easily changed.
M_{break}	Critical temperature T_c	The temperature at which the entropy phase transition occurs.
F17	Latent heat L	Hidden energy discontinuity at a first-order transition.
Stored energy	Excess free energy Delta-F	Energy above the critical manifold. Any injection = contamination.
Pass rate	Phase diagram scan	Maps entropy across all temperatures. 100% = everywhere critical.
LEGITIMATE	At criticality	Thermodynamic equilibrium. Maximum stability. RG fixed point.
FABRICATED	Off-critical	Not at equilibrium. Wants to evolve: fuse (light nuclei) or fission (heavy).
Resolution boundary	Thermometer range limit	Where the calorimeter can no longer distinguish signal from noise.
Rayleigh envelope	Incoherent energy response	Rise-peak-decay under progressive contamination = noise + coherent signal statistics.

A.5 Key Theorems & Properties

Theorem 1: Entropy invariance

For a legitimate compositional time series on S^D , $H\text{-bar}(M)$ is invariant (within threshold epsilon) under geometric-mean block decimation for all M in $[2, \text{floor}(N/5)]$.

Theorem 2: σ_A^2 predicts M_{break}

The Aitchison variance σ_A^2 of the full composition predicts the EITT break point M_{break} . Higher σ_A^2 implies earlier M_{break} . This is not empirical — it follows from how block-averaging interacts with the variation matrix.

Theorem 3: Geometric mean preserves Aitchison geometry

The geometric mean is the Frechet mean in Aitchison distance. Block-averaging with geometric means preserves the log-ratio structure of the simplex. Arithmetic means do not.

Theorem 4: F17 normalisation

The raw F17 contamination signal C_{geom} scales with σ_A^2 . Normalising F17 by σ_A^2 yields a scale-invariant alarm. An absolute threshold would destroy legitimate high-variance datasets.

Property: Perturbation isometry

$d_A(x \text{ (circleplus)} z, y \text{ (circleplus)} z) = d_A(x, y)$. Translation on the simplex preserves distances. Verified to machine precision.

Property: Powering scaling

$d_A(\alpha \text{ (circledot)} x, \alpha \text{ (circledot)} y) = |\alpha| \cdot d_A(x, y)$. Scalar multiplication scales distances. Verified to machine precision.

Property: CLR orthogonality

$\text{sum}_i \text{clr}(x)_i = 0$ for all x . The CLR maps to a (D-1)-dimensional hyperplane in R^D .

A.6 References

[1] Aitchison, J. (1986). *The Statistical Analysis of Compositional Data*. Chapman & Hall.

- [2] Pawlowsky-Glahn, V., Egozcue, J.J. & Tolosana-Delgado, R. (2015). *Modeling and Analysis of Compositional Data*. Wiley.
- [3] Egozcue, J.J., Pawlowsky-Glahn, V., Mateu-Figueras, G. & Barcelo-Vidal, C. (2003). Isometric logratio transformations for compositional data analysis. *Mathematical Geology*, 35(3), 279-300.
- [4] Shannon, C.E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3), 379-423.
- [5] Kadanoff, L.P. (1966). Scaling Laws for Ising Models Near T_c . *Physics*, 2(6), 263-272.
- [6] Wilson, K.G. (1971). Renormalization Group and Critical Phenomena. *Physical Review B*, 4(9), 3174.
- [7] Wilson, K.G. (1975). The renormalization group: Critical phenomena and the Kondo problem. *Reviews of Modern Physics*, 47(4), 773.
- [8] Wick, G.C. (1954). Properties of Bethe-Salpeter Wave Functions. *Physical Review*, 96(4), 1124.
- [9] Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *A&A*, 641, A6.
- [10] Weizsaecker, C.F. von (1935). Zur Theorie der Kernmassen. *Zeitschrift fuer Physik*, 96, 431-458.
- [11] Bethe, H.A. & Bacher, R.F. (1936). Nuclear Physics A. *Reviews of Modern Physics*, 8, 82-229.
- [12] Wang, M. et al. (2021). The AME 2020 atomic mass evaluation (II). *Chinese Physics C*, 45(3), 030003.
- [13] Mayer, M.G. (1949). On Closed Shells in Nuclei. II. *Physical Review*, 75(12), 1969.
- [14] Lunney, D., Pearson, J.M. & Thibault, C. (2003). Recent trends in the determination of nuclear masses. *Reviews of Modern Physics*, 75(3), 1021.
- [15] Butterworth, S. (1930). On the Theory of Filter Amplifiers. *Experimental Wireless*, 7, 536-541.

End of Appendix. This document is self-contained.